Power Series and Interval of Convergence

Section 1: Finding the Radius of Convergence

1 Find the radius of convergence R for the power series $\sum_{n=0}^{\infty} \frac{(x-3)^n}{4^n}$.

a. $R = 4$

b. $R = 1$

c. $R = 3$

d. $R = \infty$

2 What is the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$?

a. $R = \infty$

b. $R = 1$

c. $R = 0$

d. $R = e$

3 Find the radius of convergence of $\sum_{n=1}^{\infty} n!\, x^n$.

a. $R = 0$ (converges only at $x = 0$)

b. $R = 1$

c. $R = \infty$

d. $R = \frac{1}{e}$

Section 2: Testing Interval Endpoints

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4 Find the complete interval of convergence for $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n}$.

a. $[1, 3)$

b. $(1, 3)$

c. $[1, 3]$

d. $(1, 3]$

5 What is the interval of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n^2}$?

a. $[-1, 1]$

b. $(-1, 1)$

c. $[-1, 1)$

d. $(-1, 1]$

6 Why must endpoints always be tested individually when finding the interval of convergence?

a. The Ratio Test yields $L = 1$ at the boundary points $|x - c| = R$ and is inconclusive

b. Endpoints always diverge

c. Endpoints always converge

d. To find the center of the series

Section 3: Term-by-Term Differentiation of Power Series

7 If $f(x) = \sum_{n=1}^{\infty} c_n x^n$ has radius of convergence R, what is the radius of convergence of its derivative $f'(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}$?

a. R (the radius is unchanged)

b. $R - 1$

c. $\frac{R}{2}$

d. ∞

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8 Starting from $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, find a power series representation for $\frac{1}{(1-x)^2}$.

a. $\sum_{n=1}^{\infty} n x^{n-1}$

b. $\sum_{n=0}^{\infty} x^{2n}$

c. $\sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}$

d. $\sum_{n=1}^{\infty} n^2 x^n$

9 Can endpoint convergence change when a power series is differentiated term-by-term?

a. Yes, differentiation can lose convergence at an endpoint

b. No, endpoints never change

c. Differentiation always adds endpoints

d. Only if $R = \infty$

Section 4: Term-by-Term Integration of Power Series

10 Integrating $\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$ term-by-term yields the Maclaurin series for which function?

a. $\ln(1+x)$

b. $\arctan x$

c. e^{-x}

d. $\frac{1}{(1+x)^2}$

11 Integrating $\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$ term-by-term yields the Maclaurin series for:

a. $\arctan x$

b. $\ln(1+x^2)$

c. $\arcsin x$

d. $\cos x$

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12 Evaluate $\int_0^{0.1} \frac{1}{1+x^4} dx$ using the first two non-zero terms of its power series.

a. $0.1 - \frac{0.1^5}{5}$

b. $0.1 + \frac{0.1^5}{5}$

c. $0.1 - 0.1^4$

d. 0.1

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